

Lab: Planning and Deep Temporal Models

Learning Goal: Trace policy evaluation, analyze habit formation, and explore hierarchical planning.

Part 1: Policy Evaluation

Scenario: An agent in a 2-state world (left, right) with 2 actions (go_left, go_right) and time horizon $T=2$. Food is on the right.

Policies:

- $\pi_1 = [\text{go_right}, \text{go_right}]$
- $\pi_2 = [\text{go_left}, \text{go_right}]$
- $\pi_3 = [\text{go_left}, \text{go_left}]$

$C = [\text{left: } -1, \text{right: } +3]$

1. For each policy, trace the expected trajectory (what state at each time step)
2. Compute pragmatic value for each policy
3. Rank the policies. Which has lowest EFE?

Write your response here

Part 2: Habit Formation Dynamics

Learning Goal: See how policy priors $P(\pi)$ strengthen with experience.

Exercise: An agent has 3 policies. Track how $P(\pi)$ evolves:

Phase	$P(\pi_1)$	$P(\pi_2)$	$P(\pi_3)$	Behavior Type
Day 1 (no experience)	0.33	0.33	0.33	Goal-directed
After 10 successes with π_1				
After 100 successes with π_1				
After 1000 successes with π_1				

At what point does behavior become "habitual"? How can you tell?

Write your response here

Part 3: Deep Temporal Model Design

Learning Goal: Build a 2-level hierarchical planning model.

Scenario: Planning a trip to a conference.

Level	Timescale	States	Actions	Goal
High	Days	{home, airport, conference, hotel}	{travel, attend, rest}	Attend and present
Low	Hours	{packing, driving, checking_in, presenting}	{pack, drive, register, speak}	Execute high-level plan

1. How does the high level set goals for the low level?
2. What happens if the low level encounters an unexpected obstacle (flight cancelled)?
3. How does prediction error propagate between levels?

Write your response here

Part 4: The Complete Algorithm Trace

Learning Goal: Walk through the full Active Inference algorithm for one time step.

Setup: Simple 2-state, 2-action POMDP. Current beliefs: $q(s) = [0.3, 0.7]$. Observation: o_1 .

Trace each step:

1. **Observe:** o_1 received
2. **Infer states:** How does $q(s)$ update? (given A matrix)
3. **Evaluate policies:** Compute $G(\pi)$ for 2 policies
4. **Select policy:** Apply softmax with $\gamma = 1$
5. **Act:** Which action is executed?
6. **Learn:** What parameters are updated?

Show your reasoning at each step (use simple placeholder numbers if needed).

Write your response here

Part 5: Reflection

In 150 words, reflect: Active Inference unifies perception, action, learning, communication, and planning under one mathematical framework. Is this unification a strength (elegant and powerful) or a weakness (too abstract, loses meaningful distinctions)? What does unification gain us as scientists?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Policy evaluation	EFE computation and ranking
2	Dynamics analysis	Habit formation and policy priors
3	Hierarchical design	Deep temporal models
4	Algorithm tracing	Complete Active Inference loop
5	Theoretical reflection	Value of theoretical unification